

Hybrid Model Predictive and Nonlinear Feedback Control for Chaos Suppression and MPPT Enhancement in DFIG-Based Wind Energy Systems

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Abstract—Doubly fed induction generator (DFIG)-based wind turbines are strongly nonlinear electromechanical systems that can exhibit chaotic dynamics under perturbed parameters and adverse operating conditions, leading to degraded power quality and compromised stability. This paper analyses a third-order rotor-current/speed model of the DFIG with a physically consistent parameter set, characterizes its equilibria, dissipativity and Lyapunov exponents, and proposes a hybrid controller acting only on the two rotor voltages. An outer speed loop generates the torque-current reference, nonlinear feedback compensation cancels the coupled current dynamics, and a constrained receding-horizon Model Predictive Control (MPC) with integral action, whose terminal weight and gain are obtained from a linear matrix inequality, handles the converter voltage limits. Simulations show that the controller suppresses the chaotic oscillations within 0.13 s without steady-state error, and tracks Maximum Power Point Tracking (MPPT) references under turbulent wind with a speed RMS error of 0.008 rad/s and a stator-power RMS error below 1 % of the mean power.

Index Terms—Doubly-Fed Induction Generator (DFIG), Wind Energy Systems, Chaotic Dynamics, Nonlinear Dynamics, Maximum Power Point Tracking (MPPT), Hybrid Control, Model Predictive Control (MPC), Nonlinear Feedback Control.

I. INTRODUCTION

Wind energy has become an important part of the transition toward lower-carbon electricity generation [1]– [3]. In 2024, global wind capacity increased by about 117 GW, bringing the installed capacity to roughly 1136 GW [4]– [6].

The doubly fed induction generator (DFIG) is widely used in large onshore wind turbines. Since the rotor is connected through a partial-scale back-to-back converter, typically rated at about 25–30 % of the machine power, torque and active/reactive power can be controlled over a relatively wide speed range [7]– [10]. Nevertheless, the nonlinear machine equations and the coupling between electrical and mechanical variables may produce pronounced transients. Under sufficiently severe operating conditions, bifurcations and chaotic responses may also appear [11]– [15]. Grid disturbances, turbulent wind, machine detuning and converter saturation amplify these phenomena, which cause power oscillations and

torque ripple and conflict with grid-code requirements such as low-voltage ride-through [16]– [20].

The considered variable-speed DFIG wind turbine is described by a nonlinear model involving the two rotor-current components and the rotor speed. The model is first examined for the parameter set used in the simulations, including its equilibrium points and chaotic response. A hybrid controller based on Model Predictive Control (MPC) and nonlinear feedback compensation is subsequently used for both chaos suppression and MPPT reference tracking.

II. WIND TURBINE MODELING

The aerodynamic power captured by a horizontal-axis wind turbine can be written as [21]– [23]

$$P_a = \frac{1}{2} \pi \rho R^2 C_p(\lambda) v^3 \quad (1)$$

where ρ is the air density, R is the rotor radius, v is the wind speed, and $C_p(\lambda)$ is the power coefficient, which depends on the tip-speed ratio λ .

The tip-speed ratio is defined by

$$\lambda = \frac{R\omega_r}{v} \quad (2)$$

with ω_r the rotor (mechanical) angular speed.

The corresponding aerodynamic torque applied to the shaft is obtained from

$$T_a = \frac{P_a}{\omega_r} = \frac{1}{2} \pi \rho R^2 C_p(\lambda) \frac{v^3}{\omega_r} \quad (3)$$

The rotational dynamics of the wind turbine-generator set are described by

$$J \frac{d\Omega_m}{dt} = T_a - T_e - D\Omega_m \quad (4)$$

where J is the total inertia, Ω_m is the mechanical rotor speed, T_e is the electromagnetic torque produced by the generator, and D denotes the viscous friction coefficient.

Relating mechanical and electrical rotor speeds by

$$\omega_r = n_p \Omega_m \quad (5)$$

with n_p the number of pole pairs, and using $\frac{d\Omega_m}{dt} = \frac{1}{n_p} \frac{d\omega_r}{dt}$, (4) can be rewritten in electrical speed form as

$$\frac{J}{n_p} \frac{d\omega_r}{dt} = T_a - T_e - \frac{D}{n_p} \omega_r \quad (6)$$

For a DFIG expressed in the stator dq reference frame, the electromagnetic torque is given by

$$T_e = \frac{3}{2} n_p (\psi_{ds} i_{qs} - \psi_{qs} i_{ds}) \quad (7)$$

where ψ_{ds}, ψ_{qs} are the stator flux components and i_{ds}, i_{qs} are the corresponding stator current components.

Substituting (7) into (6) yields the final electromechanical dynamic equation of the rotor:

$$\frac{J}{n_p} \frac{d\omega_r}{dt} = T_a - \frac{3}{2} n_p (\psi_{ds} i_{qs} - \psi_{qs} i_{ds}) - \frac{D}{n_p} \omega_r \quad (8)$$

III. CHAOTIC DOUBLY-FED INDUCTION GENERATOR

According to (8) and the stator-flux/rotor-current coupling of the DFIG, a third-order state equation with i_{dr} , i_{qr} , and ω_r as state variables can be obtained:

$$\begin{aligned} \dot{i}_{dr} &= \left(-\frac{R_s L_m^2}{\sigma L_s^2 L_r} - \frac{R_r}{\sigma L_r} \right) i_{dr} + (\omega_s - \omega_r) i_{qr} \\ &\quad - \frac{\omega_r L_m \psi_s}{\sigma L_r L_s} + \left(-\frac{L_m}{\sigma L_s L_r} \right) u_{ds} + \frac{1}{\sigma L_r} u_{dr} \\ \dot{i}_{qr} &= -(\omega_s - \omega_r) i_{dr} + \left(-\frac{R_s L_m^2}{\sigma L_s^2 L_r} - \frac{R_r}{\sigma L_r} \right) i_{qr} \\ &\quad + \frac{R_s L_m \psi_s}{\sigma L_s^2 L_r} + \left(-\frac{L_m}{\sigma L_s L_r} \right) u_{qs} + \frac{1}{\sigma L_r} u_{qr} \\ \dot{\omega}_r &= \left(\frac{n_p}{J} \right) \left(\frac{3 n_p L_m \psi_s}{2 L_s} i_{dr} - \frac{D}{n_p} \omega_r - T_L \right) \end{aligned} \quad (9)$$

We now rewrite (9) by introducing the following state variables:

$$x_1 = i_{dr}, \quad x_2 = i_{qr}, \quad x_3 = \omega_r.$$

Let the parameters $a_1, a_2, \dots, a_8$ be defined accordingly. The resulting nonlinear system can then be expressed in the compact state-space form

$$\begin{cases} \dot{x}_1 = a_1 x_1 + (\omega_s - x_3) x_2 + a_2 x_3 + (a_3 u_{ds} + a_4 u_{dr}), \\ \dot{x}_2 = -(\omega_s - x_3) x_1 + a_1 x_2 + a_5 + (a_3 u_{qs} + a_4 u_{qr}), \\ \dot{x}_3 = a_6 x_1 - a_7 x_3 - a_8. \end{cases} \quad (10)$$

The coefficients appearing in (10) are given by

$$\begin{aligned} a_1 &= -\frac{R_s L_m^2}{\sigma L_s^2 L_r} - \frac{R_r}{\sigma L_r}, & a_2 &= -\frac{L_m \psi_s}{\sigma L_r L_s}, & a_3 &= -\frac{L_m}{\sigma L_s L_r}, \\ a_4 &= \frac{1}{\sigma L_r}, & a_5 &= \frac{R_s L_m \psi_s}{\sigma L_s^2 L_r}, & a_6 &= \frac{3 n_p^2 L_m \psi_s}{2 J L_s}, \\ a_7 &= \frac{D}{J}, & a_8 &= \frac{n_p}{J} T_L, \end{aligned}$$

with $\sigma = 1 - L_m^2 / (L_s L_r)$. The stator flux is aligned with the q axis, so that $u_{ds} = -\omega_s \psi_s$ and $u_{qs} = 0$.

The parameters listed in Table I are physically consistent: $L_s, L_r > L_m$ and $\sigma = 0.0645 \in (0, 1)$. They give $a_1 =$

TABLE I
MAIN PARAMETERS OF THE CHAOTIC DFIG SYSTEM

Parameter	Value	Parameter	Value
R_s	$5.0536 \times 10^{-5} \Omega$	n_p	2
R_r	$1.8781 \times 10^{-4} \Omega$	J	$31.18 \text{ kg} \cdot \text{m}^2$
L_s	0.0833 H	D	$71.85 \text{ N} \cdot \text{m} \cdot \text{s/rad}$
L_r	0.0844 H	T_L	$-11337 \text{ N} \cdot \text{m}$
L_m	0.0811 H	u_{dr}	2.539 V
σ	0.0645	u_{qr}	-2.936 V
ω_s	$100\pi \text{ rad/s}$	u_{ds}	-563.4 V
ψ_s	1.7933 Wb	u_{qs}	0 V

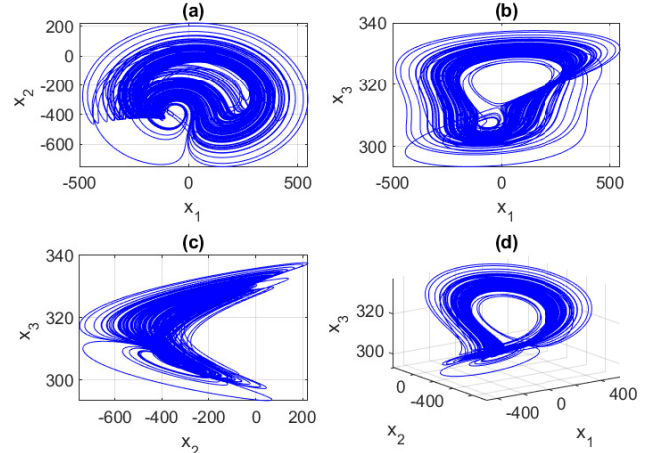


Fig. 1. The chaotic attractors: a) The projection of the chaotic attractor in plan $x_1 - x_2$; b) the projection of the chaotic attractor in plan $x_1 - x_3$; c) the projection of the chaotic attractor in plan $x_2 - x_3$; d) 3D trajectory of the system.

-0.0433 , $a_2 = -320.83$, $a_3 = -178.91$, $a_4 = 183.76$, $a_5 = 0.1946$, $a_6 = 0.3360$, $a_7 = 2.3044$, $a_8 = -727.20$. The constant terms of (10) are therefore $a_3 u_{ds} + a_4 u_{dr} = 1.0126 \times 10^5$ and $a_5 + a_3 u_{qs} + a_4 u_{qr} = -539.32$.

System (10) has three unstable equilibria: the saddle-foci $E_1 = (-64.94, -379.12, 306.11)$ (eigenvalues $1.462 \pm 7.673j$, -5.315) and $E_3 = (55.50, -272.00, 323.67)$ ($0.547 \pm 10.511j$, -3.486), and the saddle $E_2 = (-9.97, -12442.5, 314.12)$ (-65.00 , 62.65 , -0.043). The divergence $\nabla \cdot f = 2a_1 - a_7 = -2.391$ is negative, so the system is dissipative.

The chaotic attractor obtained from $x(0) = (0.3, 0.6, 3.6)^T$ is shown in Fig. 1.

In Fig. 1, the trajectory winds around the two saddle-foci E_1 and E_3 . The rotor speed remains positive and close to synchronism ($296 \leq \omega_r \leq 336 \text{ rad/s}$), while the rotor currents oscillate irregularly between about -670 and $+470$ A. The Lyapunov exponents, computed with the QR-based variational method, are $(0.202, -0.010, -2.582)$. Their sum equals the divergence -2.391 , and the Kaplan–Yorke dimension is 2.08. The positive largest exponent confirms chaos.

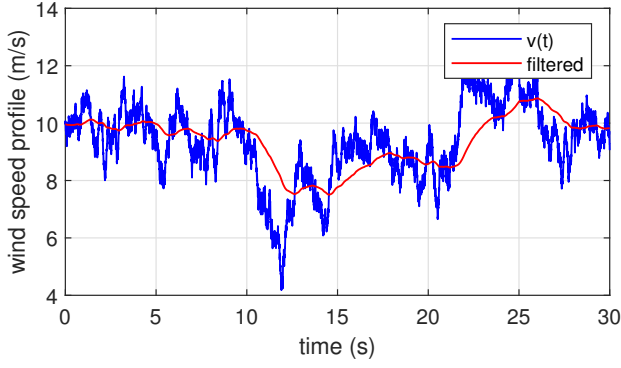


Fig. 2. Wind-speed trajectory.

IV. REFERENCE TRAJECTORY GENERATION AND TRACKING ERROR FORMULATION

The wind speed combines a mean value, low-frequency harmonics and a stochastic turbulence term:

$$v(t) = V_{mean} + \sum_{k=1}^3 A_k \sin(2\pi f_k t) + v_{turb}(t) \quad (11)$$

here V_{mean} denotes the average wind speed, A_k and f_k are the amplitudes and frequencies of the harmonic components, and $v_{turb}(t)$ is a smoothed Gaussian noise process representing atmospheric turbulence.

The resulting profile is shown in Fig. 2. The power reference is

$$P_{ref}(t) = \frac{1}{2} \rho \pi R^2 C_p v^3(t) \quad (12)$$

with C_p the optimal power coefficient. With the stator flux aligned with the q axis (Section III), the electromagnetic torque and the stator active power are governed by the d -axis rotor current, leading to the reference

$$i_{dr,ref}(t) = -\frac{2}{3} \frac{P_{ref}(t)}{V_s} \frac{L_s}{L_m} \quad (13)$$

The corresponding measured active power is

$$P_m(t) = -\frac{3}{2} \frac{L_m}{L_s} V_s i_{dr}(t) \quad (14)$$

The reference for the q -axis rotor current is derived from the stator reactive-power expression under stator-flux orientation,

$$i_{qr,ref}(t) = -\frac{2L_s}{3L_m V_s} \left(Q_s(t) - \frac{3V_s^2}{2\omega_s L_s} \right) \quad (15)$$

and, by imposing $Q_s(t) = 0$, simplifies to the constant operating point

$$i_{qr,ref} = \frac{V_s}{\omega_s L_m} = \frac{\psi_s}{L_m} \quad (16)$$

which ensures unity power-factor operation.

The reference rotor speed is obtained from the optimal tip-speed ratio, giving

$$\omega_{r,ref}(t) = \frac{G \lambda_{opt}}{R} v(t) \quad (17)$$

The complete reference trajectory is therefore

$$x_r(t) = \begin{bmatrix} i_{dr,ref}(t) \\ i_{qr,ref}(t) \\ \omega_{r,ref}(t) \end{bmatrix},$$

with tracking error $e(t) = x(t) - x_r(t)$.

V. HYBRID CONTROL OF A NOVEL CHAOTIC DFIG-BASED SYSTEM

The controlled system takes the form

$$\begin{aligned} \dot{x}_1 &= a_1 x_1 + (\omega_s - x_3) x_2 + a_2 x_3 + a_3 u_{ds} + a_4 u_1, \\ \dot{x}_2 &= -(\omega_s - x_3) x_1 + a_1 x_2 + a_5 + a_3 u_{qs} + a_4 u_2, \\ \dot{x}_3 &= a_6 x_1 - a_7 x_3 - a_8, \end{aligned} \quad (18)$$

where $x = [x_1, x_2, x_3]^T = [i_{dr}, i_{qr}, \omega_r]^T$ and the two control inputs $u_1 = u_{dr}$ and $u_2 = u_{qr}$ are the rotor voltages imposed by the rotor-side converter. The speed equation has no direct input: ω_r is controlled through the electromagnetic torque, i.e., through i_{dr} . With the parameters of Table I, (18) reads

$$\begin{aligned} \dot{x}_1 &= -0.0433 x_1 + (\omega_s - x_3) x_2 - 320.83 x_3 \\ &\quad + 1.0079 \times 10^5 + 183.76 u_1, \\ \dot{x}_2 &= -(\omega_s - x_3) x_1 - 0.0433 x_2 + 0.1946 + 183.76 u_2, \\ \dot{x}_3 &= 0.3360 x_1 - 2.3044 x_3 + 727.20. \end{aligned}$$

In open loop ($u_1 = 2.539$ V, $u_2 = -2.936$ V), (18) reduces to the chaotic system (10). The control objective is to make the state follow the reference trajectory introduced in Section IV. Consequently, convergence of the tracking error also removes the chaotic motion and provides the required MPPT behavior.

The control structure is shown in Fig. 3. Predictive control [24]–[28] is simple to configure and has a low computational cost, which makes it widely used for chaotic systems, while feedback control [29]–[32] stabilizes unstable chaotic dynamics in real time and can steer trajectories toward desired states. The proposed structure uses the nonlinear compensation to deal with the state coupling, while the predictive part accounts for the actuator constraint.

A. Outer Speed Loop

Since $T_e \propto i_{dr}$, the speed is controlled through the reference of i_{dr} . With $e_\omega = \omega_r - \omega_{r,ref}$ and the nominal model,

$$\dot{i}_{dr}^* = \frac{\dot{\omega}_{r,ref} + a_7 \omega_{r,ref} + a_8 - \lambda e_\omega - \lambda_I \int e_\omega dt}{a_6}, \quad (19)$$

where $a_8 = n_p T_L / J$ uses the load (or aerodynamic) torque. Substituting (19) into the third line of (18) gives

$$\dot{e}_\omega = -(\lambda + a_7) e_\omega - \lambda_I \int e_\omega dt + a_6 (i_{dr} - i_{dr}^*),$$

an exponentially stable linear system driven by the inner tracking error; hence $e_\omega \rightarrow 0$ once $i_{dr} \rightarrow i_{dr}^*$ (cascade argument). The inner reference is $x_d = [i_{dr}^*, i_{qr,ref}]^T$ with $i_{qr,ref} = \psi_s / L_m$ from (16).

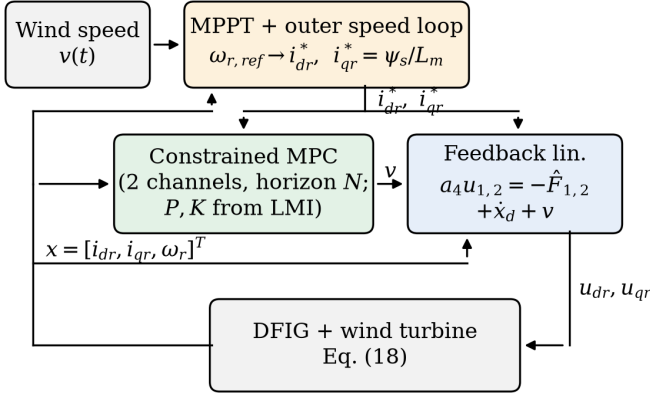


Fig. 3. Proposed hybrid MPC-nonlinear feedback control architecture: outer speed loop and inner current loops acting on the rotor voltages $u_1 = u_{dr}$, $u_2 = u_{qr}$.

B. Inner Loop: Nonlinear Feedback Compensation

Write the first two lines of (18) as $\dot{x}_{1,2} = F_{1,2}(x) + a_4 u_{1,2}$ and let $\hat{F}_{1,2}$ be the nominal model. The control law

$$u_{1,2} = \frac{1}{a_4} \left(-\hat{F}_{1,2}(x) + \dot{x}_d + v \right) \quad (20)$$

gives $\dot{e} = v + \Delta$, with $e = x_{1,2} - x_d$ and $\Delta = F_{1,2} - \hat{F}_{1,2}$ ($\Delta = 0$ nominally). The bilinear couplings $(\omega_s - x_3)x_{1,2}$ responsible for chaos are cancelled, leaving two decoupled integrators driven by v . Combining such a compensation with MPC has been shown effective for MPPT of wind turbines [33].

C. Model Predictive Control

For each channel $i \in \{1, 2\}$, the error is augmented with its integral, $\xi_i = [e_i, \int e_i dt]^T$, and sampled with period T_s :

$$\xi_i(k+1) = A\xi_i(k) + Bv_i(k), \quad A = \begin{bmatrix} 1 & 0 \\ T_s & 1 \end{bmatrix}, \quad B = \begin{bmatrix} T_s \\ T_s^2/2 \end{bmatrix}. \quad (21)$$

At each instant k , the MPC solves the quadratic program

$$\begin{aligned} \min_{v_i(\cdot|k)} & \sum_{j=1}^N \|\xi_i(k+j|k)\|_Q^2 + \|\xi_i(k+N|k)\|_P^2 \\ & + r \sum_{j=0}^{N-1} v_i(k+j|k)^2 \\ \text{s.t. (21),} & \quad |u_i(k+j|k)| \leq \bar{u}, \end{aligned} \quad (22)$$

with prediction horizon N , weights $Q = Q^T > 0$, $r > 0$, and the converter voltage limit $\bar{u}$ (via (20), a bound on v_i). Only the first optimal move is applied and the problem is solved again at $k+1$ (receding horizon) [34]; the integral is frozen when the limit is reached.

Theorem 1: Let $X = X^T > 0$ and Y satisfy the LMI

$$\begin{bmatrix} X & (AX + BY)^T & XQ^{1/2} & \sqrt{r}Y^T \\ AX + BY & X & 0 & 0 \\ Q^{1/2}X & 0 & I & 0 \\ \sqrt{r}Y & 0 & 0 & 1 \end{bmatrix} \succeq 0, \quad (23)$$

and set $P = X^{-1}$, $K = YX^{-1}$. Then $(A + BK)^T P(A + BK) - P + Q + rK^T K \preceq 0$, and if (22), augmented with the terminal constraint $\xi_i(k+N|k) \in \Omega = \{\xi^T P \xi \leq \alpha\}$ (α such that $K\xi$ respects the limits in Ω), is feasible at $k=0$ with $\Delta = 0$, it remains feasible and $\xi_i = 0$ is asymptotically stable.

Proof: The Schur complement of (23) gives $X - (AX + BY)^T X^{-1}(AX + BY) - XQX - rY^T Y \succeq 0$; pre- and post-multiplying by P with $Y = KX$ yields the stated inequality. Thus $\xi^T P \xi$ is a Lyapunov function for $v = K\xi$ and Ω is invariant. By the standard terminal-cost argument [34], the optimal cost decreases by at least $\|\xi_i(k)\|_Q^2 + rv_i(k)^2$ at each step, which proves recursive feasibility and asymptotic stability. ■

Remark: The LMI formulation of the terminal ingredients follows [35]. The maximal-volume solution of (23) is the Riccati solution, so without active constraints the MPC returns exactly $v_i = K\xi_i$: a single gain $K = [k_e, k_I]$ from the LMI replaces the previous matrices K_1 , K_2 and the diagonal K .

VI. NUMERICAL SIMULATIONS

All simulations integrate (18) with a fourth-order Runge-Kutta scheme (step $T_s/4$). The controller uses $T_s = 1$ ms, $N = 10$, $Q = \text{diag}(1, 3)$, $r = 10^{-6}$ (currents scaled by 100 A), which gives $K = [-619.2, -1069.5]$ and an LMI residual below 10^{-15} ; the outer loop uses $\lambda = 10 \text{ s}^{-1}$, $\lambda_I = 25 \text{ s}^{-2}$, and the rotor voltages are limited to $|u_{dr}|, |u_{qr}| \leq 200$ V.

A. Chaos Suppression

The system starts from $x(0) = (0.3, 0.6, 3.6)^T$ with the open-loop voltages of Table I and evolves on the chaotic attractor. At $t = 10$ s the controller is activated with the target $\omega_r^* = 1.05\omega_s = 329.9 \text{ rad/s}$ and $i_{qr}^* = \psi_s/L_m = 22.1 \text{ A}$ ($Q_s = 0$); the corresponding equilibrium $i_{dr}^* = 98.0 \text{ A}$ is held with $u_{dr} = 29.3 \text{ V}$ and $u_{qr} = -8.4 \text{ V}$.

Fig. 4 shows that the chaotic oscillations disappear as soon as the controller is activated: the speed reaches the positive super-synchronous target and the rotor currents settle at their equilibrium values, using only the rotor voltages. Table II shows that the speed enters a 1% band of the target in 0.13 s without steady-state error. During the first instants the rotor voltages reach the 200 V limit, which the MPC accounts for explicitly; afterwards they stay below 34 V and settle at $u_{dr} = 29.3 \text{ V}$, $u_{qr} = -8.4 \text{ V}$.

B. MPPT Tracking

For normal operation, the turbine torque $T_L = -T_a(v, \omega_r)$ of (3) replaces the constant load, with $R = 20 \text{ m}$, $\lambda_{opt} = 8.1$, $C_{p,\max} = 0.48$ and $G = 86$ in (17) (pole pairs included), and the damping is set to its nominal value $D = 0.1 \text{ N}\cdot\text{m}\cdot\text{s/rad}$. The

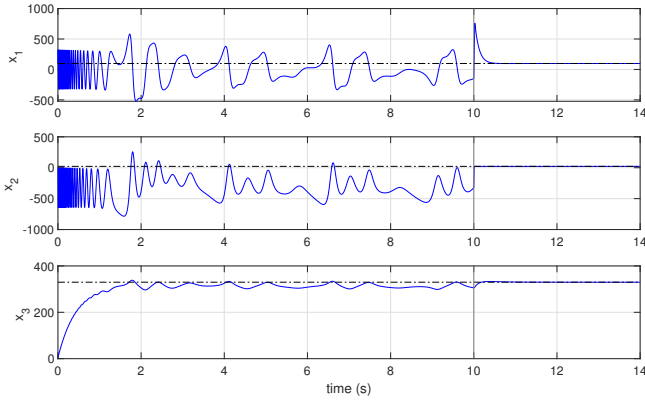


Fig. 4. Time evolution of the state variables before and after control activation at $t = 10$ s (dash-dot: target).

TABLE II
CHAOS SUPPRESSION PERFORMANCE (FL-MPC)

Metric	Value
IAE of the speed error	3.63 rad
Settling time (1 % band)	0.13 s
Final speed error	1.8×10^{-6} rad/s
Steady rotor voltages u_{dr}, u_{qr}	29.3 V, -8.4 V

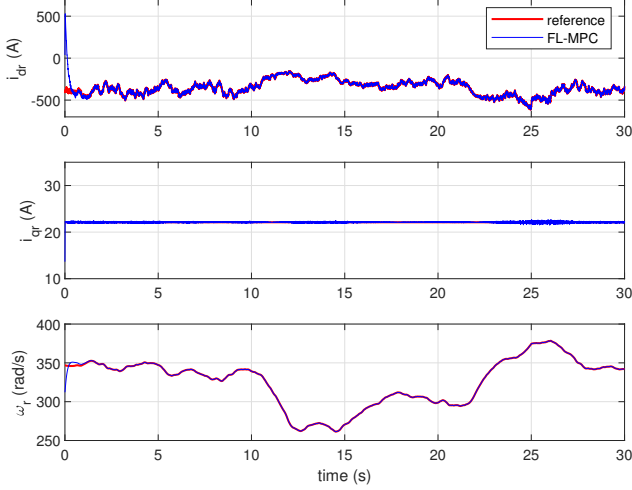


Fig. 5. Tracking performance of the state variables i_{dr} , i_{qr} , and ω_r under the proposed hybrid control scheme.

machine is driven by the wind of Fig. 2 (mean 9 m/s); $\omega_{r,ref}$ follows the low-pass filtered wind (time constant 2 s), so that $\omega_{r,ref}$ stays between 279 and 378 rad/s and $P_{s,ref}$ between 142 and 478 kW. The results in Figs. 5 and 6 indicate that the three controlled states and the stator active power remain close to their MPPT references despite the imposed wind variations. After a short initial transient, the rotor speed converges to its reference and follows it throughout the simulation.

VII. CONCLUSION

The nonlinear behavior of the considered DFIG wind energy conversion system was examined using the selected machine

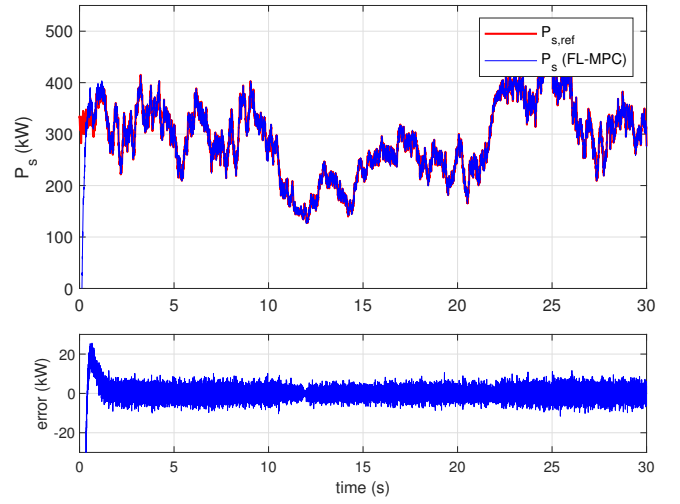


Fig. 6. Active power tracking performance: stator power P_s and MPPT reference $P_{s,ref}$, and power tracking error.

TABLE III
MPPT TRACKING PERFORMANCE OF FL-MPC ($t \geq 5$ s)

Metric	Value
RMS error of ω_r	0.008 rad/s
RMS error of i_{dr}	3.28 A
RMS error of i_{qr}	0.085 A
RMS error of P_s	2.70 kW (0.93 % of mean P_s)
Speed settling time (2 % band)	0.13 s

parameters. The equilibrium points, dissipativity condition, and Lyapunov exponents characterize the chaotic operating regime obtained in open loop. A hybrid controller using only the rotor voltages was then introduced. The outer loop provides the i_{dr} reference from the speed error, while nonlinear feedback compensation reduces the effect of the current-state coupling. The MPC part deals with the rotor-voltage constraint and uses the LMI-based terminal ingredients described above. In simulation, the chaotic oscillations are removed after controller activation, while the MPPT case shows satisfactory stator-power tracking under turbulent wind conditions. Extension of the model to include converter and grid dynamics, together with experimental validation, remains to be considered.

CONFLICT OF INTEREST

The authors declare there are no conflicts of interest.

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